Treatment of contaminated MSS with PAA

Matthias Rüdt, MSAT Process Engineer DSP

Treatment of contaminated MSS with PAA

Background

PAA study results

Strategy for MSS resin decontamination

Project Goal and Scope

- Goal:
  - Identify approaches for decontaminating microbiologically contaminated MabSelect SuRe Resin
- Scope:
  - 7 contaminated resin lots stored at Basel DS manufacturing
- Drivers:
  - Supply chain limitations: MSS is currently difficult to procure and essential to process
  - Economic: Total value 14 Mio. CHF
- Identified methods:
  - Treatment with peracetic acid (PAA) → **prioritized method**
  - Radiation sterilization (gamma radiation)

Overview resins

LOT	Bioburden / (CFU/mL)	Observed germs	Remaining cycles
1	250	Microbacterium sp, Lysinibacillus sp, Paenibacillus sp	235
2	<10	Microbacterium sp, Bacillus sp, Paenibacillus sp	234
3	2100	P. glucanolyticus	210
4	190	P. glucanolyticus	222
5	700	P. glucanolyticus	150
6	tbd	tbd	211
7	tbd	tbd	225

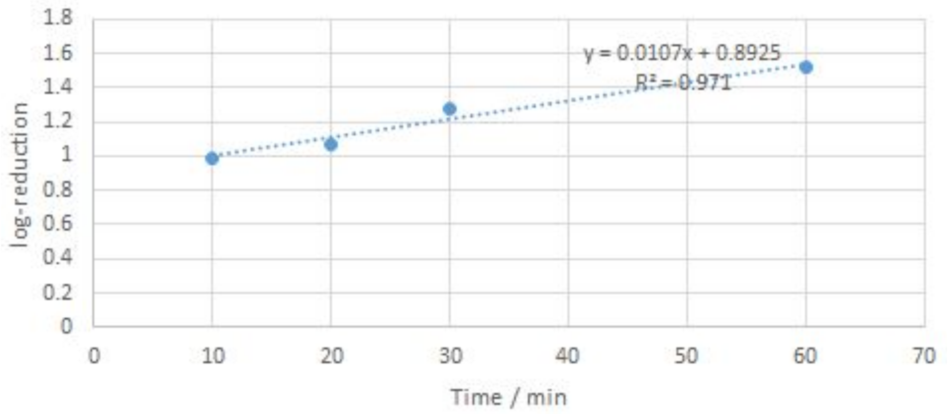
Treatment of contaminated MSS with PAA

Background

PAA study results

Strategy for MSS resin decontamination

20 mM PAA study with P. glucanolyticus spiking



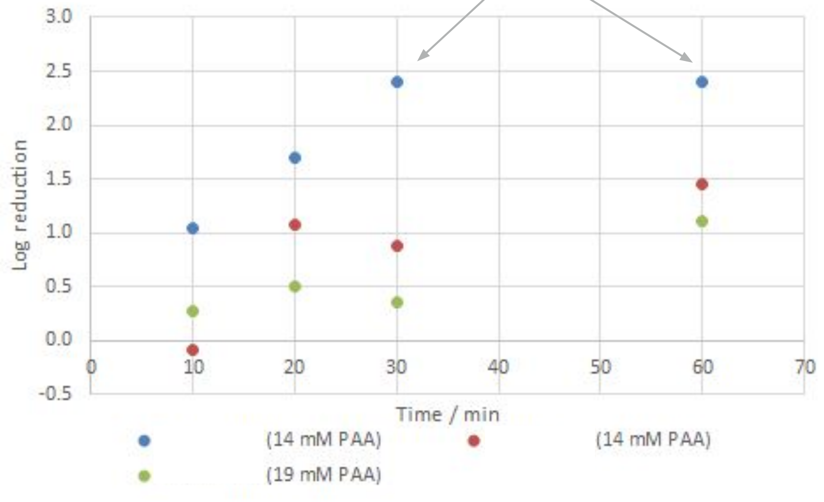
Experimental conditions:

- 30% resin slurry spiked with P. glucanolyticus spores
- No flow
- Effective PAA concentration ~14 mM (diluted due to water in resin)
- PAA stock solution 15% PAA

Study by Johanna Schenker & Andreas Krug

20 mM PAA study with contaminated resin lots

<LOQ (mixture of organisms)



Experimental conditions:

- 30% resin slurry
- No flow
- No spiking
- PAA concentration ~14 mM (diluted due to water in resin), resp. ~19 mM
- PAA stock solution 36.7% PAA (Acros)

Study by Johanna Schenker & Andreas Krug

Long-exposure time study with PAA and PAB

P. glucanolyticus spores w/o resin

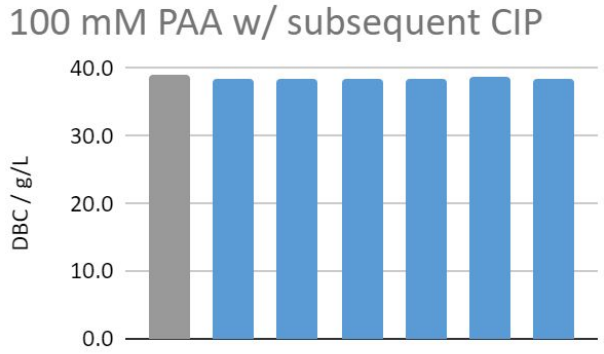
- Goal: get insight into ‘long-term’ dynamics of PAA treatment and comparison PAB efficacy

	Log-Reduction	
Timepoint / h	**PAA (20 mM)**	**PAB**
6	3.5	-1.9
24	3.7	-1.8
48	4.7	-2.0

- Conclusion:
  - 20 mM PAA results in increased reduction over prolonged incubation periods.
  - PAB potentially splits up spore clusters (??)

100 mM AET PAA study + Binding capacity measurements

Exposure / min	Log-reduction
15	>6.5
30	>6.5
60	>6.5



Treatment of contaminated MSS with PAA

Background

PAA study results

Strategy for MSS resin decontamination

MabSelect SuRe decontamination - general outline

FMEA

MSS decontamination

1. Patient safety assessment due to microbial remnants

- **Goal**: exclude impact due to microbial remnants on resin
- **Method**: mathematical quantification

2. Microbiological efficacy of decontamination

- **Goal**: assure necessary microbial depletion
- **Method**: AET study and extrapolation

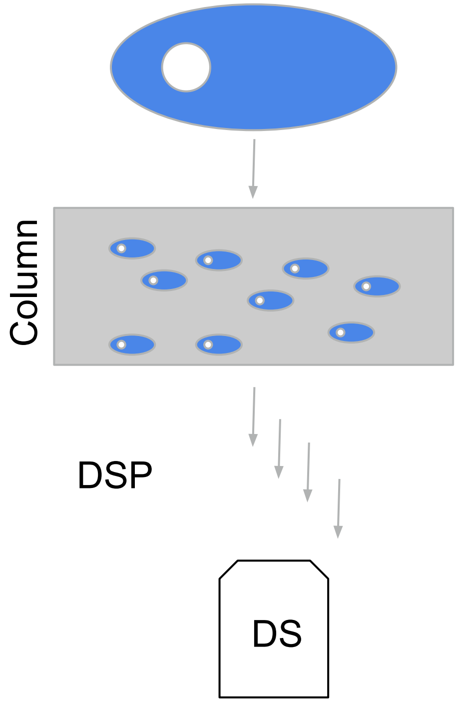
3. Resin performance after decontamination

- **Goal**: ensure resin performance over lifetime
- **Method**: Resin lifetime study after decontamination + testing of individual resins

1. Patient safety assessment

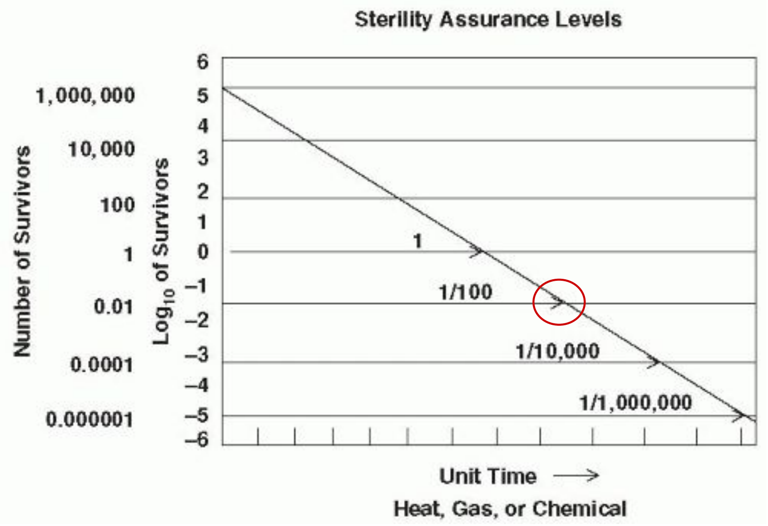
General approach:

1. Calculate mass of critical components per cell for different species.
2. Calculate the total amount of critical components based on the amount of resin used in the process and the measured contamination level (in CFU/mL).
3. Deplete over DSP with expected DSP performance per critical component.
4. Compare concentration of critical components in DS with Maximum Allowable Concentration in final drug as per internal SOP



2. AET study

- Spiking studies with *P. glucanolyticus* spores, used MSS resin and 100 mM PAA
- Sampling points at 15 min, 30 min and 1 hour
- Extrapolation to achieve sterility on column scale (226 L)
- Observed spore depletion: >6.6 log-level reduction for all test points.



3. Resin lifetime study with naive MSS

Procedure:

1. Treat resin with 100 mM PAA (3 h treatment) + neutralize
2. 50 h of PAB incubation + neutralize
3. Cycle until before end of lifetime (249 cycles)
   a. Every 25 cycles, sample pool for full analytics
4. Treat with 100 mM PAA (3h treatment) + incubation
5. 50 h of PAB incubation + neutralize
6. Perform last cycle + full analytics
7. End of lifetime carry over sample
8. 400 h of PAB incubation + DBC measurement

Analyzed metrics:

Similar but extension of original resin lifetime study:

- Performance indicator: Yield, pool pH, chromatogram evaluation
- Process-related contaminants: CHO DNA, CHO HCP, Luterol, leached protein A
- Oxidized product variants

3. Linking study: resin performance after decontamination

Procedure:

- Each resin lot to be decontaminated at-scale is first decontaminated in small scale
- Testing of resin performance with full analytics for 1 cycle to demonstrate comparability to resin lifetime study

Analyzed metrics:

The same as during resin lifetime study:

- Performance indicator: Yield, pool pH, chromatogram evaluation
- Process-related contaminants: CHO DNA, CHO HCP, Luterol, leached protein A
- Oxidized product variants

Further steps & open questions

- Resin lifetime study has started.
- Most likely, we will try to file the decontamination procedure.
- Any feedback from industry:
  - Similar approaches?
  - Differences?

Acknowledgement

- Penzberg Microbial QC
  - Andreas Krug
  - Johanna Schenker
- MSAT Basel
  - Dominik Hediger
  - Laura Schöpflin
  - Andreas Gebert
  - Eike Dümmer
  - Robert Puskeiler
- Global
  - Mike Knight
  - Friedrich von Wintzingerode

Doing now what patients need next